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Foreword: Using Bronfenbrenner's ecological systems theory to frame quantitative, qualitative, and mixed research

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Teddlie and Tashakkori (2010) identified the following nine important issues or controversies in contemporary mixed research: (a) conceptual stances; (b) the conceptual/methodological/methods interface; (c) research question or research problem; (d) language; (e) design issues; (f) analysis issues; (g) drawing inferences; (h) practical applications (e.g., pedagogy, collaboration, and other models, funding); and (i) cross-disciplinary and cross-cultural applications. According to Teddlie and Tashakkori, 'there are obvious overlaps across some of these areas' (p. 12). Indeed, we believe that these nine issues/controversies can be simplified by characterizing them as representing one or more of the following three elements: Philosophy (e.g., conceptual stances in mixed research; the conceptual/methodological/methods interface in mixed research), policy (e.g., the language of mixed research), and practice (e.g., research question or problem; design issues in mixed research; analysis issues in mixed research; practical issues in the application of mixed research). By *philosophy*, we are referring to what Greene (2006, 2008) refer to as *philosophical assumptions and stances*, which are the core epistemological assumptions of the methodology, or what Greene (2008) refers to as *mental models*, which represent:

> the set of assumptions, understandings, predispositions, and values and beliefs with which all social inquirers approach their work. Mental models influence how we craft our work in terms of what we choose to study and how we observe and listen, what we see and hear, what we interpret as salient and important, and indeed what we learn from our empirical work. (p. 12)

By *policy*, we are referring to what Greene (2006, 2008) labeled as *sociopolitical commitments*, which represents specification and justification of how the research is located in society. Finally, by *practice*, we mean both what Greene (2006, 2008) called *inquiry logics*, which situates the researcher in the investigation in a manner that the phenomenon of interest is observed, documented, and understood or explained in defensible ways; as well as what Greene (2006, 2008) referred to as *guidelines for research practice*, which provide specific techniques for research practice such as the sampling design (e.g., sampling scheme, sample size), research design, data collection, and data analysis.

This mapping of Teddlie and Tashakkori's (2010) nine important issues or controversies in contemporary mixed research onto the three elements of philosophy, policy, and practice should help to explain our choice of title for this special issue, namely, *Mixed Methods Research: Philosophy, Policy, and Practice in Education*. Indeed, all three elements are represented by two or more of the seven special issue articles. Specifically, in this special issue, two of the articles represent philosophy, two articles represent policy, and three articles represent practice. With respect to philosophy, the two articles are those by Jean-Luc Patry and by Thomas W. Christ. In his thought-provoking article entitled, 'Beyond multiple methods: critical multiplism on all levels', Jean-Luc Patry discusses the evolution of the concept of critical multiplism, which provided a justification for the use of multiple research approaches and to which mixed research belongs. Jean-Luc Patry identifies five levels of research-related statements (i.e., theory of science, object theory, methods, data analysis, and research ethics) and then applied critical multiplism to each level. Finally, Jean-Luc Patry illustrates that the theory of situation specificity (i.e., different behavior of the same person in different situations) can serve as a theoretical lens for

critical multiplism and that it can help address several problems that come to the fore in research. So, as can be seen, Jean-Luc Patry's article clearly represents the area of philosophy.

Thomas W. Christ's article entitled, 'Paradigm considerations and mixed methods in social science research', is the second article that represents the area of philosophy. In his excellent article, Thomas Christ provides the latest discussion about alternatives to pragmatism – particularly the research philosophy called critical realism, which is based on the premise that there are various levels of reality, which range from the objective that is independent of human understanding, to subjective truths that humans understand and interpret in the process of meaning making (Christ, 2010). Christ also presents a tool, which he calls the *worldview matrix*, to help researchers, methodology instructors, and students identify the epistemology, ontology, axiology, and methodology pertaining to their mixed research studies.

The two articles that represent the area of policy are those by Jerome De Lisle and by Mogenthiran S. Pillay, Brigitte Smit, and Coert Loock. In his intriguing article entitled, 'Exploring the value of integrated findings in a multiphase mixed methods evaluation of the continuous assessment program in the Republic of Trinidad and Tobago', Jerome De Lisle uses mixed research techniques to conduct a theory-driven evaluation of a continuous assessment program (i.e., CAP; an externally directed, curriculum-based assessment schemes used by classroom teachers for both summative and formative purposes) in the Republic of Trinidad and Tobago. These researchers focused their report on the construction and the use of integrated findings, which they contended have great policy relevance.

In their well-designed study entitled, 'Policy disjuncture Between the National Curriculum Statement (NCS) and Curriculum 2005 Training Initiatives', Mogenthiran S. Pillay, Brigitte Smit, and Coert Loock used mixed research techniques to investigate the effectiveness of training for the NCS against the recommendations of the Curriculum Review Committee (CRC) in South Africa. Further, they explored the possibility of policy disjuncture between the training for the NCS with that of the Curriculum 2005. Also, these researchers investigated whether the recommendations of the CRC were essential and, if so, whether they had improved the training according to teachers' perceptions. Their research, which represents the first comparative study of training initiatives and the first study to examine the recommendations of the CRC, clearly has policy implications for the training of teachers in South Africa.

The three articles that represent the area of practice are those by David Plowright; by Sandra M. Harris; and by Renata Clerici, Giancarlo Gola, and Emanuela Cisco. In the interesting article entitled, 'To what extent do postgraduate students understand the principles of mixed methods in educational research?', David Plowright reported the findings of an exploratory mixed research study examining postgraduate students' understandings of the use of mixed research. He combined data extracted from an on-line questionnaire completed by postgraduate students (Stage 1) with small-group, face-to-face discussions of participants selected from the first stage (i.e., Stage 2). Because the researcher examined understandings of the *use* of mixed research, this study clearly represents the area of practice. However, due to the fact that David Plowright also examined postgraduate students' views about the underlying methodological and philosophical principles associated with conducting research in general and mixed research in particular, this study also represents the area of philosophy.

In her excellent study entitled, 'Development of the Perceptions of Mentoring Relationships Survey (PMRS): a mixed methods approach', Sandra M. Harris used mixed research techniques to develop and to construct-validate a quantitative measure of perceptions of the mentoring relationship. In the first phase, this researcher used open-ended questions to obtain participants' perceptions of the roles in mentoring relationships of mentors and mentees, as well as their perceptions of the benefits of mentoring relationships. In the second phase, Sandra Harris used findings from the first phase to develop the instrument and then to score-validate it. Because

this scale subsequently can be used to ascertain perceptions of mentoring relationships, this study clearly represents the area of practice.

Finally, in their well-designed study entitled, 'Qual–quant analysis of the statistical content in Italian primary school general books', Renata Clerici, Giancarlo Gola, and Emanuela Cisco utilized mixed research techniques to analyze a random sample of fourth- and fifth-grade Italian primary mathematics general books. A content analysis of these textbooks led to codes that were analyzed statistically and that yielded a content typology. Through the use of textbooks by teachers in practice, this study contributes to this issue's focus in the area of practice.

Thus, as can be seen, this special issue contains articles that represent almost an even distribution with respect to emphasis on philosophy, policy, and practice, thereby justifying the title of this special issue. The distribution of articles on topics in international contexts also makes the special issue useful for a diverse audience. Also, as a set, we believe that these articles could serve as a useful complement to the readings assigned by an instructor of a course on mixed research.

As co-editors and first-round copyeditors of *Research in the Schools* (Anthony J. Onwuegbuzie and Rebecca K. Frels), former editor of *Educational Researcher* (Anthony J. Onwuegbuzie), guest editor of previous *International Journal of Multiple Research Approaches* special issues (Anthony J. Onwuegbuzie, Kathleen M. T. Collins, and Rebecca K. Frels), editorial assistant/production editor of *Research in the Schools* (Rebecca K. Frels), and reviewers for and editorial board members of numerous journals, we have had the opportunity to review hundreds of quantitative, qualitative, and mixed research articles over the years. And this experience has led us to identify a theoretical framework that incorporates virtually all research that represents the social, behavioral, and health sciences. Specifically, we arrived at the conclusion that philosophy, policy, and practice might be linked and viewed through Bronfenbrenner's ecological systems theory, due to the fact that ecological theory is about relationships in each of these genres. Thus, we conceptualize that Bronfenbrenner's (1979) ecological systems theory can be mapped onto the quantitative, qualitative, and mixed research process. In particular, Bronfenbrenner's (1979) theory can be used to identify and to plan research studies representing all three traditions across the social, behavioral, and health sciences. In what follows, we outline our conceptualization.

Bronfenbrenner's ecological systems model

Bronfenbrenner's (1979) ecological systems model consists of four environmental levels – the microsystem, the mesosystem, the exosystem, and the macrosystem – with each level impacting differently the development of each person. According to Bronfenbrenner, the microsystem, which represents Level 1, involves the immediate environment with which the child/adolescent closely interacts such as the classroom, playground, recreation center, home, friend's home, neighborhood, and religious institution. Formally, Bronfenbrenner (1979) defined it as follows: 'a microsystem is a pattern of activities, roles, and interpersonal relations experienced by the developing person in a given setting with particular physical and material characteristics' (p. 22).

The mesosystem, which represents Level 2, was defined by Bronfenbrenner (1979) as 'the interrelations among two or more settings in which the developing person actively participates (such as, for a child, the relations among home, school, and neighborhood peer group: For an adult, among family, work, and social life)' (p. 25). In other words, the mesosystem refers to relations among microsystems or connections among contexts such as the relationship between family experiences and school experiences, between school experiences and neighborhood experiences, and between family experiences and peer experiences. For instance, children who are bullied at school (school experiences) might withdraw from their parents (family experiences) at home.

The exosystem, which represents Level 3, refers to 'one or more settings that do not involve the developing person as an active participant, but in which events occur that affect or are affected by, what happens in the setting containing the developing person' (Bronfenbrenner, 1979, p. 25). That is, the exosystem characterizes links between

a social setting in which the person does not have an active role and the person's immediate context. For example, a person's experience at home might be influenced by his/her spouse's/partner's experiences at work – a newborn baby might reduce the numbers of hours slept by the mother's/father's spouse/partner, which, in turn, might affect the productivity levels of the spouse/partner at work, which, in turn, might affect the relationship between the spouse/partner and his/her supervisor and/or colleagues, which, in turn, might affect the relationship between the couple at home.

Finally, the macrosystem, which represents Level 4, according to Bronfenbrenner (1979), refers to:

> consistencies in the form and content of lower-order systems (micro-, meso-, and exo-) that exists, or could exist, at the level of the subculture or the culture as a whole, along with any belief systems or ideology underlying such inconsistencies. (p. 26)

Alternatively stated, the macrosystem, which is the highest level, involves the larger cultural context (e.g., society, community) surrounding the person that includes societal belief systems, cultural norms, ideologies, policies, or laws that indirectly influence the person. Members of a cultural group share a common identity, heritage, and values. Macrosystems are developed temporally, meaning that they evolve over time.

We conceptualize that the four levels comprising Bronfenbrenner's (1979) ecological systems model could be mapped onto the research process representing all three research traditions, namely, qualitative, quantitative, and mixed research, and it is applicable across the social, behavioral, and health fields. Specifically, we refer to: (a) *micro-research studies* (i.e., Level 1) as involving research whereby one or more persons, groups, or other living organisms that are studied within his/her/their/its immediate environment[s]; (b) *meso-research studies* (i.e., Level 2) as involving research whereby one or more persons, groups, or other living organisms are investigated within other systems in which the he/she/they/it spends time; (c) *exo-research studies* (i.e., Level 3) as involving research whereby one or more persons, groups, or other living organisms are examined within systems by which he/she/they/it might be influenced but of which he/she/they/it does not play an active role; and (d) *macro-research studies* (i.e., Level 4), as involving research whereby one or more persons, groups, or other living organisms are studied within the larger cultural world or society surrounding him/her/them/it. Each of these four levels is illustrated in Figure 1.

Although, qualitative, quantitative, or mixed research studies could be conducted at one of these four levels (i.e., micro-research,

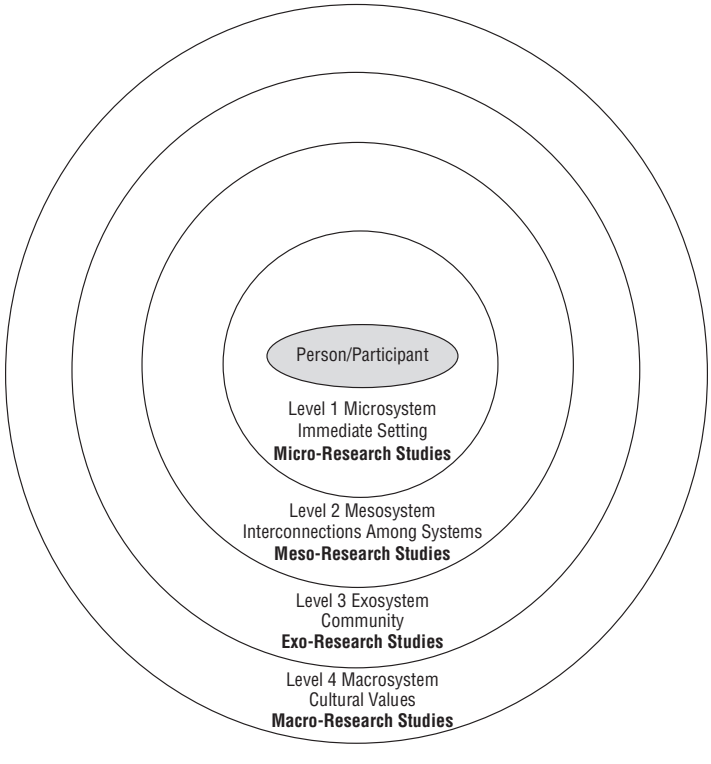

Figure 1: A visual representation of Bronfenbrenner's (1979) ecological systems model and levels of research

meso-research, exo-research, macro-research), in the applications of multimethod research studies (i.e., using multiple research methods within the same tradition [i.e., quantitative *or* qualitative research]), typically only one level is involved. In mixed research studies, it is possible for two or more levels to be involved, with the quantitative and qualitative phase(s) conducted at the same or at different levels. An important implication of our conceptualization for research in general is that just as Bronfenbrenner's (1979) four levels are interconnected, so too are qualitative, quantitative, and mixed research studies that involve an examination of the same phenomenon at the same or different levels.

Our four-level conceptualization has important implications for generalization (e.g., policy, practice) because it helps the researcher bound the inquiry or conceptual/theoretical framework with respect to the generalizability[1] of the findings, concept, model, or theory. For a given empirical study, the appropriate type of generalization not only depends on the sampling design (e.g., type of sampling schemes [i.e., purposive vs. random], sampling scheme [e.g., stratified random sampling, cluster sampling, convenience sampling, criterion sampling; cf. Collins, 2010; Kemper, Stringfield, & Teddlie, 2003; Onwuegbuzie & Collins, 2007; Teddlie & Yu, 2007], sample size, subsample size[s], group size[s] per approach, number of observational units per participant) but it also depends on the level of each research phase inherent in the study (i.e., micro-research, meso-research, exo-research, macro-research). Accordingly, researchers can make *within-level generalization and*/or *across-level generalization*, wherein within-level generalization represents one of Onwuegbuzie, Slate, Leech, and Collins's (2009) five types of generalizations (e.g., external statistical generalization, analytic generalization[1]) of the findings, concept, or theory being made within the same level (i.e., micro, meso, exo, macro) and across-level generalization, which occurs in mixed research studies, wherein one of the types of generalizations identified by Onwuegbuzie et al. (2009), and selected by the researcher to generalize the findings, concept(s), or theory made at one research level (e.g., Level 1) is combined or mixed with one of the types of generalizations of the findings, concept, or theory being made at another research level (e.g., Level 4) to yield meta-inferences (i.e., inferences made from qualitative and quantitative findings, concept, or theory being integrated into either a coherent whole or two distinct sets of coherent wholes; cf. Tashakkori & Teddlie, 1998).

Application of the model to examples of generalizations

Our seven special issue articles could be subdivided into empirical studies (N = 5) and conceptual/theoretical works (N = 2). Of the five empirical studies, four of them represent macro-research (i.e., Level 4), namely, the studies of Jerome De Lisle; David Plowright; Renata Clerici, Giancarlo Gola, and Emanuela Cisco; and Mogenthiran S. Pillay, Brigitte Smit, and Coert Loock. Specifically, Jerome De Lisle combined analytic generalizations from findings emerging at Phase 1 (i.e., involving assessment

[1] Onwuegbuzie, Slate, Leech, and Collins (2009) have identified five major types of generalizations that researchers can make, as follows: (a) *external (statistical) generalizations* (i.e., making generalizations, inferences, or predictions on data obtained from a representative statistical [i.e., optimally random] sample to the *population* from which the sample was drawn); (b) *internal (statistical) generalizations* (i.e., making generalizations, inferences, or predictions on data obtained from one or more representative or elite participants [e.g., key informants, politically important cases, sub-sample members]); (c) *analytic generalizations* (i.e., 'applied to wider theory on the basis of how selected cases "fit" with general constructs'; Curtis, Gesler, Smith, & Washburn, 2000, p. 1002); (d) *case-to-case transfers* (i.e., making generalizations or inferences from one case to another [similar] case [Firestone, 1993; Kennedy, 1979; Miles & Huberman, 1994]); and (e) *naturalistic generalizations* (i.e., the readers of the article make generalizations entirely, or at least in part, from their personal or vicarious experiences [Stake, 2005], such that meanings arise from personal experience, and are adapted and reified by repeated encounter [Stake, 1980; Stake & Trumbull, 1982]). Virtually all quantitative, qualitative, and mixed research studies involve the use of one or more of these five types of generalizations.

of the context and the program theory from the stakeholders' perspectives), external statistical generalizations from findings extracted from Phase 2 (i.e., examination of frequencies for all CAP activities practiced by teachers; a rating scale on the use of different types of assessment; and number of the different types of teacher feedback), and analytic generalizations from findings emerging at Phase 3 (i.e., multisite case study based on schools with different contexts and implementation rates, thereby representing different educational subcultures in the Republic of Trinidad and Tobago) to yield Level 4 meta-inferences. David Plowright combined external statistical generalizations from findings emerging at Stage 1 (i.e., on-line questionnaire survey) and analytic generalizations[1] from findings emanating from Stage 2 (i.e., small-group, face-to-face discussions) to yield Level 4 meta-inferences through the lens of a postgraduate sub-culture. Renata Clerici, Giancarlo Gola, and Emanuela Cisco combined analytic generalizations from findings emanating from the first phase (i.e., resulting in a content analysis of textbooks) with external statistical generalizations from findings extracted at the second phase (i.e., resulting in a canonical non-linear correlation analysis of the transformed codes) to yield Level 4 meta-inferences in the educational/cultural context in Italy. Mogenthiran S. Pillay, Brigitte Smit, and Coert Loock combined external statistical generalizations from findings emerging at Phase 1 (i.e., structured questionnaire) and analytic generalizations from findings emanating from Phase 2 (i.e., focus groups) to yield Level 4 meta-inferences in the educational/cultural context based in South Africa. In these three preceding studies, the mitigating factor of culture (Level 4) influenced the context of the mixed investigation.

In contrast, one special issue study represents micro-research (i.e., Level 1), namely, the study of Sandra M. Harris, who combined analytic generalizations from findings emanating from Phase 1 (i.e., collection of qualitative data on students' perceptions of mentoring relationships) with external statistical generalizations from findings extracted at the second phase (i.e., developing and score-validating a new instrument that was based on the findings at Stage 1) to yield Level 1 meta-inferences, whereby the perceptions of students within their immediate environments formed the basis for the meta-inferences. The two special issue articles that represent conceptual/theoretical works were those by Jean-Luc Patry and Thomas W. Christ. Both of these articles outlined philosophical concepts that have important implications for research conducted at all four levels (i.e., micro-research, meso-research, exo-research, macro-research) because the researcher's philosophical stance is central towards actualizing research into practice.

As you read the articles in this special issue, we encourage you to keep in mind the philosophical, policy, and practice implications of each and the associated research level(s) pertinent for each article. It is our hope that through providing the three elements – philosophy, policy, and practice – as they align to the nine issues/controversies in mixed research with the framework of Bronfenbrenner, educational and social science researchers might continue the dialog for understanding research relationships/implications, as well as the utility and scope of mixed research.

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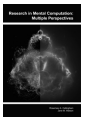

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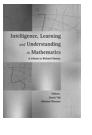

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